

1. Create the following tables with properly specifying Primary keys, Foreign keys and solve the following queries.

BRANCH(**Branchid**,Branchname,HOD)

STUDENT(**USN**,Name,Address,Branchid,sem)

BOOK(**Bookid**,Bookname,Authorid,Publisher,Branchid)

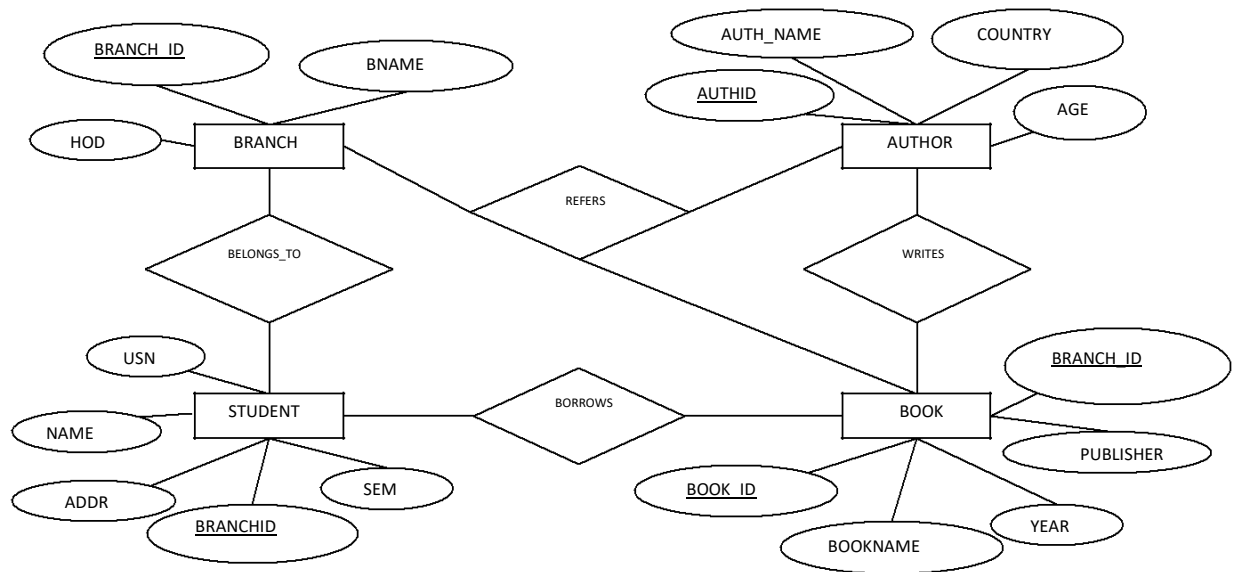
AUTHOR(**Authorid**,Authorname,Country,age)

BORROW(USN,Bookid,Borrowed_Date)

Queries:

1. List the details of Students who are all Studying in 2ndsem MCA.
2. List the students who are not borrowed any books.
3. Display the USN, Student name, Branch_name, Book_name, Author_name,Books_Borrowed_Date of 2ndsem MCA Students who borrowed books.
4. Display the number of books written by each Author.
5. Display the student details who borrowed more than two books.
6. Display the student details who borrowed books of more than one Author.
7. Display the Book names in descending order of their names.
8. List the details of students who borrowed the books which are all published by the same Publisher.

ER DIAGRAM :



SCHEMA DIAGRAM

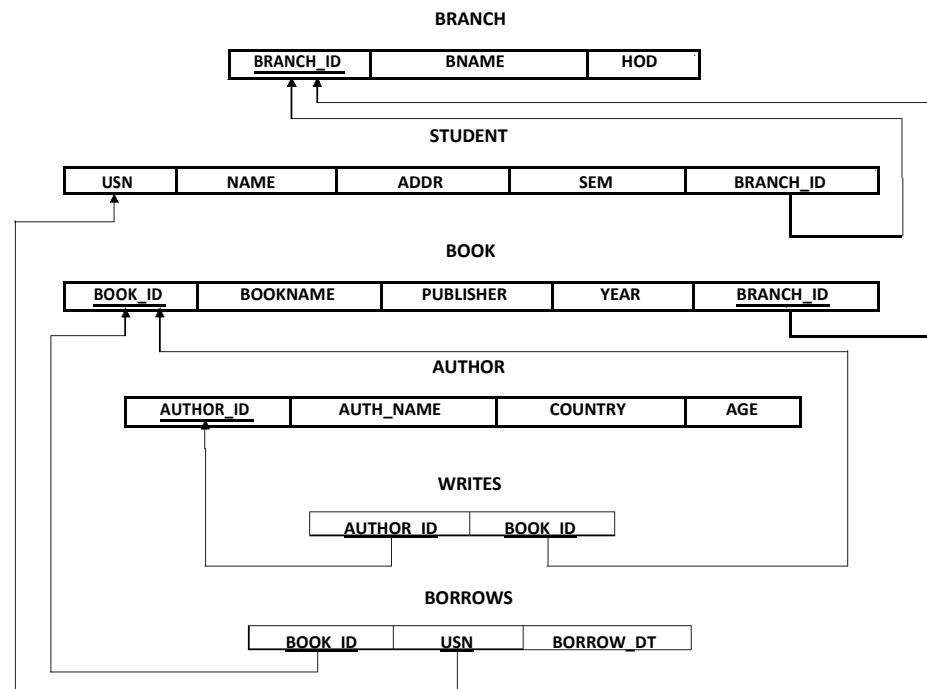


TABLE 1:BRANCH

```
create table BRANCH(BranchID varchar(20) primary key,  
  Bname varchar(20),HOD varchar(20) not null,UNIQUE(Bname));
```

TABLE CREATED.

```
insert into BRANCH values('CSE_UG','COMP_SC','H1');  
select * from BRANCH;
```

BRANCHID	BNAME	HOD
CSE_UG	COMP_SC	H1
ISE_UG	INFO_SC	H2
MCA_PG	MASTER_MCA	H3
CSE_PG	MTECH_CSE	H4

4 ROWS SELECTED.

TABLE 2:STUDENT

```
create table STUDENT(USN varchar(25) primary key, Name varchar(25) not null,  
  Addr varchar(25) not null,BranchID varchar(20) references BRANCH,Sem int);  
TABLE CREATED.
```

```
insert into STUDENT values('USN1','NAME1','ADDR1','MCA_PG','2');
```

USN	NAME	ADDR	BRANCHID	SEM
USN1	NAME1	ADDR1	MCA_PG	2
USN2	NAME2	ADDR2	MCA_PG	2
USN3	NAME3	ADDR3	CSE_UG	1
USN4	NAME4	ADDR4	ISE_UG	3

TABLE 3:BOOK

```
create table BOOK(BookID varchar(30) primary key,Bookname varchar(30) not  
  null, Publisher varchar(30) not null,BranchID varchar(20) references BRANCH,year  
  int); TABLE CREATED.  
insert into BOOK values('B1','C','P1','MCA_PG','2010');
```

```
select * from BOOK;
```

BOOKID	BOOKNAME	PUBLISHER	BRANCHID	YEAR
-----	-----	-----	-----	-----
B1	C	P1	MCA_PG	2010
B2	CPP	P2	CSE_UG	2008
B3	OS	P3	ISE_UG	2015
B4	PYTHON	P4	CSE_PG	2013
B5	SSW	P5	MCA_PG	2012
B6	JAVA	P6	CSE_PG	2008

6 rows selected

TABLE 4:AUTHOR

```
create table AUTHOR(AuthID varchar(30) primary key,  
AuthName varchar(30) not null,country varchar(30) not null,age int);
```

TABLE CREATED.

```
insert into AUTHOR values('AUTH1','A1','INDIA','30');
```

```
select * from AUTHOR;
```

AUTHID	AUTHNAME	COUNTRY	AGE
-----	-----	-----	-----
AUTH1	A1	INDIA	30
AUTH2	A2	INDIA	32
AUTH3	A3	INDIA	34
AUTH4	A4	INDIA	36

4 ROWS SELECTED.

TABLE 5:WRITES

```
create table WRITES(AuthID varchar(30) references  
AUTHOR, BookID varchar(30) references BOOK);
```

TABLE CREATED.

```
insert into WRITES values('AUTH1','B1');
```

```
select * from WRITES;
```

AUTHID	BOOKID
-----	-----
AUTH1	B1
AUTH2	B1
AUTH3	B1
AUTH2	B2
AUTH3	B2
AUTH4	B2
AUTH1	B3
AUTH3	B4
AUTH3	B5
AUTH4	B6

10 rows selected.

TABLE 6:BORROWS

```
create table BORROWS(BookID varchar(30) references BOOK,  
USN varchar(30) references STUDENT,Borrow_DT varchar(30));
```

TABLE CREATED.

```
insert into BORROWS values('B1','2BL1','20-DEC-11');  
select * from BORROWS;
```

BOOKID	USN	BORROW_DT
-----	-----	-----
B1	2BL1	20-DEC-11
B1	USN1	20-DEC-12
B2	USN2	15-AUG-13
B3	USN3	22-OCT-14
B4	USN1	15-SEP-12
B1	USN2	20-DEC-13
B2	USN1	15-AUG-13
B3	USN4	22-DEC-14

7 rows selected.

Queries:

1. List the details of Students who are all Studying in 2ndsem MCA.

Solution :

```
select * from STUDENT where Sem=2 and BranchID in(select BranchID from BRANCH
where Bname='MASTER_MCA');
```

RESULT:

USN	NAME	ADDR	BRANCHID	SEM
USN2	NAME2	A2	MCA_PG	2

2. List the students who are not borrowed any

books. Solution :

```
select * from STUDENT where USN in((select USN from STUDENT) minus (select USN from
BORROWS));
```

RESULT :

USN	NAME	ADDR	BRANCHID	SEM
USN4	NAME4	A4	MCA_PG	5

3. Display the USN, Student name, Branch_name, Book_name, Author_nameBooks_Borrowed_Date of 2ndsem MCA Students who borrowed books.
Solution :

```
select s.USN,s.NAME,B.Bname,B1.Bookname,B2.BORROW_DT,A.Authname from
STUDENT S,BRANCH B,BOOK B1,AUTHOR A,BORROWS B2,WRITES
W where B2.BookID=B1.BookID and S.BranchID=B.BranchID and
W.BookID=B1.BookID and W.AuthID=A.AuthID and
B2.BookID=W.BookID and A.AuthID=W.AuthID and
B2.BookID=B1.BookID and B2.USN=S.USN and S.USN IN(select USN from STUDENT
where Sem=2 and BranchID in(select BranchID from BRANCH where
Bname='MASTER_MCA'));
```

USN	NAME	BNAME	BOOKNAME	BORROW_DT	AUTHNAME
USN2	NAME2	MASTER_MCA	C	20-DEC-12	A1
USN2	NAME2	MASTER_MCA	C	20-DEC-12	A2
USN2	NAME2	MASTER_MCA	C	20-DEC-12	A3
USN2	NAME2	MASTER_MCA	CPP	15-AUG-13	A2
USN2	NAME2	MASTER_MCA	CPP	15-AUG-13	A3
USN2	NAME2	MASTER_MCA	CPP	15-AUG-13	A4

6 rows selected.

4. Display the number of books written by each Author. Solution :

```
select AuthID,count(*) from WRITES group by AuthID;  
RESULT :
```

AUTHID	COUNT(*)
-----	-----
AUTH1	2
AUTH2	2
AUTH3	4
AUTH4	2

5. Display the student details who borrowed more than two books.
Solution :

```
select * from STUDENT where USN in(select USN from BORROWS group by USN having  
count(*)>2);
```

RESULT ;

USN	NAME	ADDR	BRANCHID	SEM
-----	-----	-----	-----	-----
USN1	NAME1	A1	CSE_UG	1

6. Display the student details who borrowed books of more than one Author.
Solution :

```
select * from STUDENT where  
USN in(select USN from BORROWS where  
BookID in(select BookID from WRITES group by BookID having count(*)>2));
```

RESULT :

USN	NAME	ADDR	BRANCHI	SEM
-----	-----	-----	-----	-----
USN1	NAME1	A1	CSE_UG	1
USN2	NAME2	A2	MCA_PG	2

7 Display the Book names in descending order of their names.

Solution :

```
select * from BOOK order by Bookname desc;
```

RESULT :

BOOKID	BOOKNAME	PUBLISHER	BRANCHID	YEAR
B6	PYTHON	P3	CSE_PG	2008
B3	JAVA	P3	ISE_UG	2015
B5	J2EE	P2	MCA_PG	2012
B2	CPP	P2	CSE_PG	2008
B4	COBOL	P4	CSE_PG	2013
B1	C	P1	MCA_PG	2010

8 List the details of students who borrowed the books which are all published by the same Publisher.

Solution :

```
select * from STUDENT where USN in(select USN from BORROWS where  
BookID in(select BookID from BOOK where Publisher='P3'));
```

RESULT :

USN	NAME	ADDR	BRANCHID	SEM
USN3	NAME3	A3	CSE_PG	3
USN4	NAME4	A4	MCA_PG	5

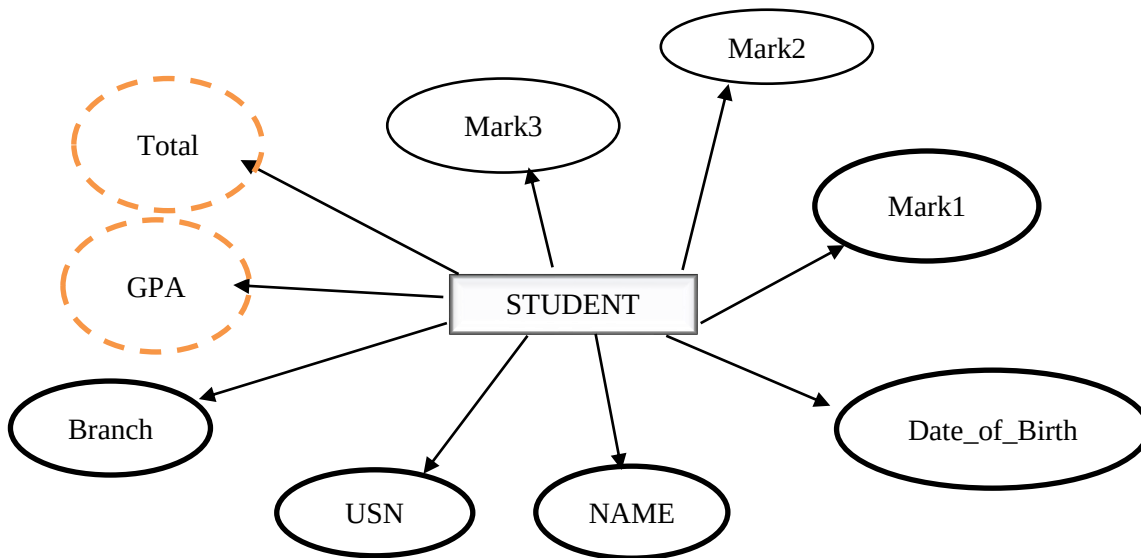
2. Consider the following Schema:

STUDENT(USN,name,date_of_birth,mark1,mark2,mark3,total,GPA)

Execute the following queries:

1. Update the column total by adding the columns mark1, mark2, mark3.
2. Find the GPA Score of all the students.
3. Find the students who born on a particular year of birth from the date-of-birth column.
4. List the students who are studying in a particular branch of study.
5. Find the maximum GPA score of the student branch-wise.
6. Find the students whose name ends with the alphabet “S”.
7. Find the students whose name ends with the alphabets “AR”.
8. Delete the student details whose USN is given as 1001.

ER-DIAGRAM



SCHEMA DIAGRAM

Student

USN	NAME	DOB	BRANCH	Mark1	Mark2	Mark3	Total	GPA
-----	------	-----	--------	-------	-------	-------	-------	-----

create table student(USN Integer Primary key,Name Varchar (25), DOB date not null,Branch Varchar (15),Mark1 Integer,Mark2 Integer,Mark3 Integer,Total Integer,GPA float);
table created

insert into student(USN,name,DOB,branch,mark1, mark2, mark3) values(03,'aishu','22-MAR-1999','cs',80,90,95);

select * from student;

USN	NAME	DOB	BRANCH	MARK1	MARK2	MARK3	TOTAL	GPA
3	aishu	22-MAR-99	cs	80	90	95		
5	ammu	03-JUN-99	IS	88	80	91		
12	laxmi	24-SEP-99	MCA	88	92	95		
22	saru	04-JUN-99	MBA	80	95	95		
2	deepa	22-MAR-99	cs	81	89	95		
9	bhimu	08-JAN-99	IS	88	70	91		
16	pavan	04-APR-99	MCA	89	92	90		
24	rohit	21-JAN-99	MBA	81	88	95		

Queries---->

1. update student set total = mark1+ mark2+ mark3;

8 rows updated.

select * from student;

USN	NAME	DOB	BRANCH	MARK1	MARK2	MARK3	TOTAL
3	aishu	22-MAR-99	cs	80	90	95	265
5	ammu	03-JUN-99	IS	88	80	91	259
12	laxmi	24-SEP-99	MCA	88	92	95	275
22	saru	04-JUN-99	MBA	80	95	95	270
2	deepa	22-MAR-99	cs	81	89	95	265
9	bhimu	08-JAN-99	IS	88	70	91	249
16	pavan	04-APR-99	MCA	89	92	90	271
24	rohit	21-JAN-99	MBA	81	88	95	264

8 rows selected.

2. update student set GPA = (total*100)/300;

8 rows updated.

select * from student;

USN	NAME	DOB	BRANCH	MARK1	MARK2	MARK3	TOTAL	GPA
3	aishu	22-MAR-99	cs	80	90	95	265	88.3333333
5	ammu	03-JUN-99	IS	88	80	91	259	86.3333333
12	laxmi	24-SEP-99	MCA	88	92	95	275	91.6666667
22	saru	04-JUN-99	MBA	80	95	95	270	90
2	deepa	22-MAR-99	cs	81	89	95	265	88.3333333
9	bhimu	08-JAN-99	IS	88	70	91	249	83
16	pavan	04-APR-99	MCA	89	92	90	271	90.3333333
24	rohit	21-JAN-99	MBA	81	88	95	264	88

8 rows selected.

3. (select USN,name,DOB,mark1,mark2,mark3,total,GPA from student where DOB='24-SEP-1999');

USN NAME	DOB	MARK1	MARK2	MARK3	TOTAL	GPA
12 laxmi	24-SEP-99	88	92	95	275	91.6666667

4. select * from student where batch = 'MBA';

USN NAME	DOB	BRANCH	MARK1	MARK2	MARK3	TOTAL	GPA
22 saru	04-JUN-99	MBA	80	95	95	270	90
24 rohit	21-JAN-99	MBA	81	88	95	264	88

5. select usn,name,branch,total,gpa from student where gpa in(select max(gpa) from student group by batch);

USN NAME	BRANCH	TOTAL	GPA
5 ammu	IS	259	86.3333333
22 saru	MBA	270	90
12 laxmi	MCA	275	91.6666667
3 aishu	cs	265	88.3333333

4 rows selected.

Select * from student order by batch, GPA;

USN NAME	DOB	BRANCH	MARK1	MARK2	MARK3	TOTAL	GPA
9 bhimu	08-JAN-99	IS	88	70	91	249	83
5 ammu	03-JUN-99	IS	88	80	91	259	86.3333333
24 rohit	21-JAN-99	MBA	81	88	95	264	88
22 saru	04-JUN-99	MBA	80	95	95	270	90
16 pavan	04-APR-99	MCA	89	92	90	271	90.3333333
12 laxmi	24-SEP-99	MCA	88	92	95	275	91.6666667
2 deepa	22-MAR-99	cs	81	89	95	265	88.3333333
3 aishu	22-MAR-99	cs	80	90	95	265	88.3333333

8 rows selected.

6. select * from student where name like 's%';

USN	NAME	DOB	BRANCH	MARK1	MARK2	MARK3	TOTAL	GPA
22	saru	04-JUN-99	MBA	80	95	95	270	90

7. select * from student where name like '%n';

USN	NAME	DOB	BRANCH	MARK1	MARK2	MARK3	TOTAL	GPA
16	pavan	04-APR-99	MCA	89	92	90	271	90.3333333

8. select * from student where USN = 05;

USN	NAME	DOB	BRANCH	MARK1	MARK2	MARK3	TOTAL	GPA
5	ammu	03-JUN-99	IS	88	80	91	259	86.3333333

(Create by view)---->

create view student_score as (select USN,name,DOB,branch,mark1,mark2,mark3, mark1+mark2+mark3 as total, total * 100/300 as GPA from MCA_PG);
View created.

select * from student_score;

USN	NAME	DOB	BRANCH	MARK1	MARK2	MARK3	TOTAL	GPA
3	aishu	22-MAR-99	cs	80	90	95	265	88.3333333
5	ammu	03-JUN-99	IS	88	80	91	259	86.3333333
12	laxmi	24-SEP-99	MCA	88	92	95	275	91.6666667
22	saru	04-JUN-99	MBA	80	95	95	270	90
2	deepa	22-MAR-99	cs	81	89	95	265	88.3333333
9	bhimu	08-JAN-99	IS	88	70	91	249	83
16	pavan	04-APR-99	MCA	89	92	90	271	90.3333333
24	rohit	21-JAN-99	MBA	81	88	95	264	88

8 rows selected.

Note: We can execute all above queries using view (i.e View Name).

Use of Between Operator

select usn,name, to_char(dob,'dd/mon/yyyy') as date_output from student where dob between to_date('22/01/1999','dd/mm/yyyy') and to_date('01/07/1999','dd/mm/yyyy');

3. Design an ER-diagram for the following scenario, Convert the same into a relational model and then solve the following queries.

Consider a Cricket Tournament “ABC CUP” organized by an organization. In the tournament there are many teams are contesting each having a Teamid,Team_Name, City, a coach. Each team is uniquely identified by using Teamid. A team can have many Players and a captain. Each player is uniquely identified by Playerid, having a Name, and multiple phone numbers,age. A player represents only one team. There are many Stadiums to conduct matches. Each stadium is identified using Stadiumid, having a stadium_name,Address (involves city,area_name,pincode).A team can play many matches. Each match played between the two teams in the scheduled date and time in the predefined Stadium. Each match is identified uniquely by using Matchid. Each match won by any of the one team that also wants to record in the database. For each match man_of_the match award given to a player.

Queries:

1. Display the youngest player (in terms of age) Name, Team name , age in which he belongs of the tournament.
2. List the details of the stadium where the maximum number of matches were played.
3. List the details of the player who is not a captain but got the man_of _match award at least in two matches.
4. Display the Team details who won the maximum matches.
5. Display the team name where all its won matches played in the same stadium.

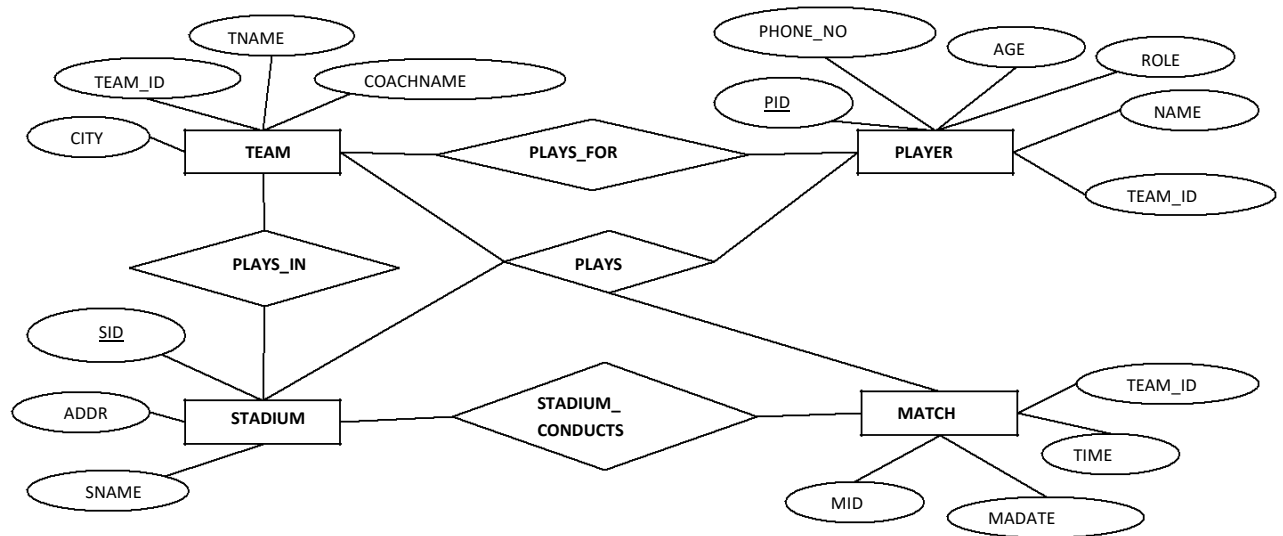
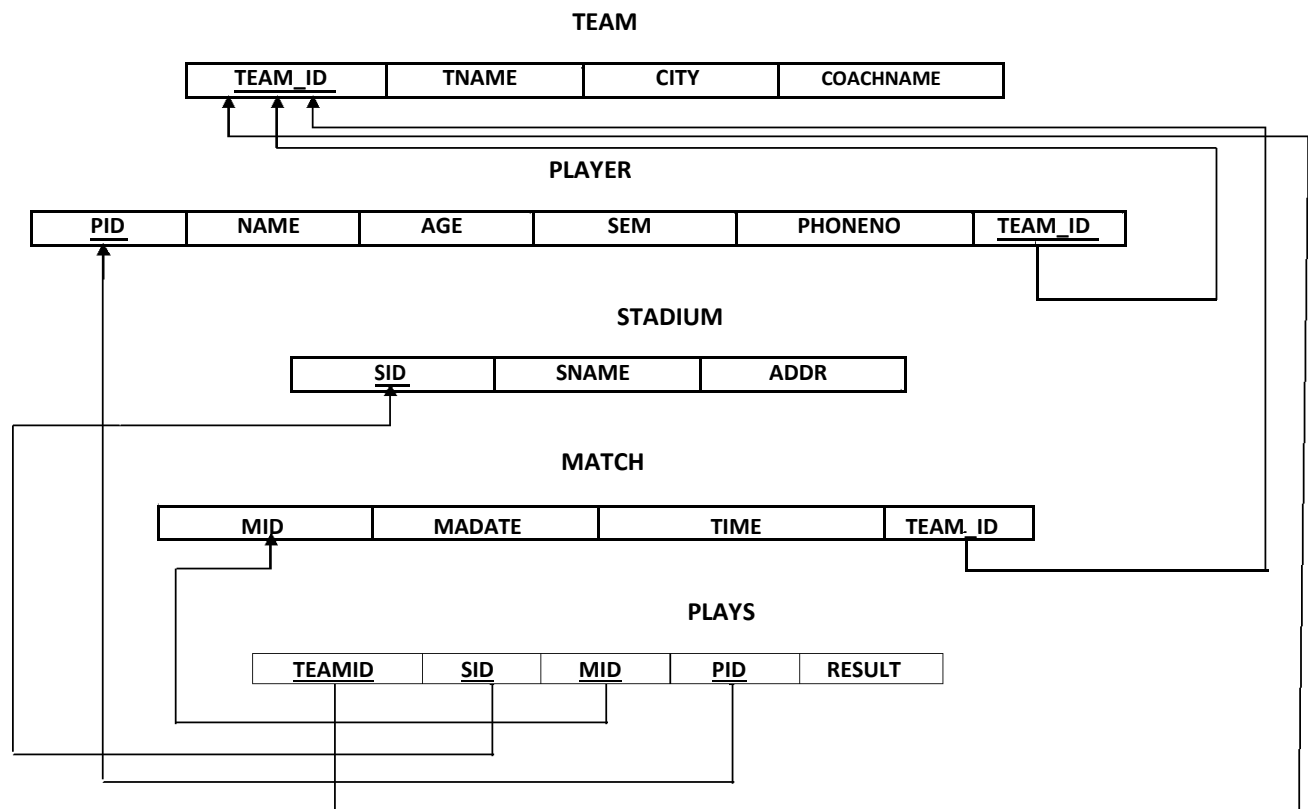
ER DIAGRAM :**SCHEMA DIAGRAM :**

TABLE 1 : TEAM

```
create table TEAM(TEAMID varchar(10) primary key,TNAME varchar(25) not null,CITY varchar(25)
not null,COACH_NAME varchar(25) not null,UNIQUE(TNAME));
```

TABLE CREATED.

```
insert into TEAM values('T1','N1','C1','CH1');
```

```
select * from TEAM;
```

TEAMID	TNAME	CITY	COACH_NAME
T1	N1	C1	CH1
T2	N2	C2	CH2
T3	N3	C3	CH3
T4	N4	C4	CH4

4 ROWS SELECTED.

.....

TABLE 2 : PLAYER

```
create table PLAYER(PID varchar(10) primary key,NAME varchar(25) not null,PHONE_NO int not
null,AGE int not null,ROL varchar(20) not null,TEAMID varchar(10) references
TEAM,UNIQUE(PHONE_NO));
```

TABLE CREATED.

```
insert into PLAYER values('P1','PNAME1','111111111','18',PLAYER,'T1');
```

```
select * from PLAYER;
```

PID	NAME	PHONE_NO	AGE ROL	TEAMID
P1	PNAME1	111111111	18 PLAYER	T1
P2	PNAME2	222222222	22 PLAYER	T2
P3	PNAME3	333333333	24 CAPTAIN	T3
P4	PNAME4	444444444	17 PLAYER	T4
P5	PNAME5	555555555	25 CAPTAIN	T2
P6	PNAME6	666666666	21 PLAYER	T1
P7	PNAME7	777777777	23 CAPTAIN	T1
P8	PNAME8	888888888	24 CAPTAIN	T4
P9	PNAME9	999999999	28 PLAYER	T3

9 rows selected.

TABLE 3 : STADIUM

```
create table STADIUM(SID varchar(10) primary key,SNAME varchar(25) not null,ADDR varchar(25)
not null);
TABLE CREATED.
```

```
insert into STADIUM values('S1','SN1','ADDR1');
```

```
select * from STADIUM;
```

SID	SNAME	ADDR
----	-----	-----
S1	SN1	ADDR1
S2	SN2	ADDR2
S3	SN3	ADDR3
S4	SN4	ADDR4

4 ROWS SELECTED.

.....

TABLE 4 : MATCH

```
create table MATCH(MID varchar(25) primary key,TEAMID varchar(10) references
TEAM,M_DATE date not null,TIME float not null);
```

TABLE CREATED.

```
insert into MATCH values('M1','T1','05-JAN-17','10');
```

```
select * from MATCH;
```

MID	TEAMID	M_DATE	TIME
-----	-----	-----	-----
M1	T1	05-JAN-17	10
M2	T2	06-JAN-17	10
M3	T3	07-JAN-17	2
M4	T4	08-JAN-17	5
M5	T2	09-JAN-17	8
M6	T3	09-JAN-17	2
M7	T1	10-JAN-17	10
M8	T4	11-JAN-17	9

8 rows selected.

.....

TABLE 5 : PLAYS

create table PLAYS(SID varchar(10) references STADIUM,MID varchar(25) references MATCH,TEAMID varchar(10) references TEAM,PID varchar(10) references PLAYER,RESULT varchar(10) not null);

TABLE CREATED.

insert into PLAYS values('S1','M1','T1','P7','WON');

select * from PLAYS;

SID	MID	TEAMID	PID	RESULT
----	-----	-----	-----	-----
S1	M1	T1	P7	WON
S2	M2	T2	P2	WON
S3	M3	T3	P3	LOSE
S4	M4	T4	P4	WON
S3	M5	T2	P5	LOSE
S2	M6	T3	P9	WON
S1	M7	T1	P3	WON
S4	M8	T3	P9	WON

8 rows selected.

.....

QUERIES:

1. Display the youngest player (in terms of age) Name, Team name , age in which he belongs of the tournament.

Solution :

select P.NAME,T.TNAME,P.AGE from PLAYER P,TEAM T where P.TEAMID=T.TEAMID and P.AGE in (select min(AGE) from PLAYER);

RESULT :

NAME	TNAME	AGE
-----	-----	-----
PNAME4	N4	17

.....

2. List the details of the stadium where the maximum number of matches were played. Solution :

```
select * from STADIUM where
```

```
SID in(select SID from PLAYS group by SID having count(*)>=(select max(count(*) from PLAYS group by SID));
```

RESULT :

SID	SNAME	ADDR
S1	SN1	ADDR1
S2	SN2	ADDR2
S3	SN3	ADDR3
S4	SN4	ADDR4

3. List the details of the player who is not a captain but got the man_of _match award at least in two matches.

Solution :

```
select P.PID,P.NAME,P.ROL from PLAYER P,PLAYS P1
```

```
where P.PID=P1.PID and P.ROL!='CAPTAIN' and P1.RESULT='WON' group by P.PID,P.NAME,P.ROL having count(*)>=1;
```

RESULT :

PID	NAME	ROLE
P2	PNAME2	PLAYER
P4	PNAME4	PLAYER
P9	PNAME9	PLAYER

.....

4. Display the Team details who won the maximum matches.

Solution :

```
select * from TEAM where
```

```
TEAMID in(select TEAMID from PLAYS where
```

```
RESULT='WON' group by TEAMID having count(*)=(select max(count(*) from PLAYS where RESULT='WON' group by TEAMID));
```

RESULT :

TEAMID	TNAME	CITY	COACH_NAME
T1	N1	C1	CH1
T3	N3	C3	CH3

.....

5. Display the team name where all its won matches played in the same stadium. Solution :

```
select TNAME from TEAM where  
TEAMID in(select TEAMID from PLAYS where  
exists(select TEAMID from PLAYS where  
RESULT ='WON' and SID in(select SID from PLAYS where RESULT='WON')));
```

RESULT :

TNAME

N1

N2

N3

N4

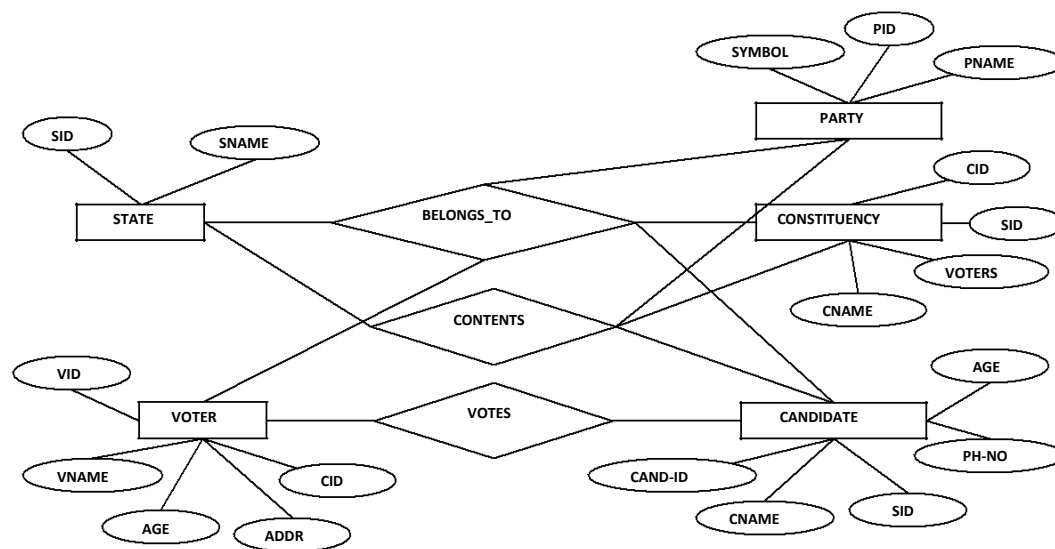
4. Design an ER-diagram for the following scenario, Convert the same into a relational model, normalize Relations into a suitable Normal form and then solve the following queries.

A country wants to conduct an election for the parliament. A country having many constituencies. Each constituency is identified uniquely by Constituency_id, having the Name, belongs to a state, Number_of_voters. A constituency can have many voters. Each voter is uniquely identified by using Voter_id, having the Name, age, address (involves Houseno,city,state,pincode). Each voter belongs to only one constituency. There are many candidates contesting in the election. Each candidates are uniquely identified by using candidate_id, having Name, phone_no, age, state. A candidate belongs to only one party. There are many parties. Each party is uniquely identified by using Party_id, having Party_Name, Party_symbol. A candidate can contest from many constituencies under a same party. A party can have many candidates contesting from different constituencies. No constituency having the candidates from the same party. A constituency can have many contesting candidates belongs to different parties. Each voter votes only one candidate of his/her constituency.

Queries:

1. List the details of the candidates who are contesting from more than one constituencies which are belongs to different states.
2. Display the state name having maximum number of constituencies.
3. Create a stored procedure to insert the tuple into the voter table by checking the voter age. If voter's age is at least 18 years old, then insert the tuple into the voter else display the "Not an eligible voter msg" .
4. Create a store procedure to display the number_of_voters in the specified constituency. Where the constituency name is passed as an argument to the stored procedure.
5. Create a TRIGGER to UPDATE the count of " Number_of_voters" of the respective constituency in "CONSTITUENCY" table , AFTER inserting a tuple into the "VOTERS" table.

ER DIAGRAM :



SCHEMA DIAGRAM :

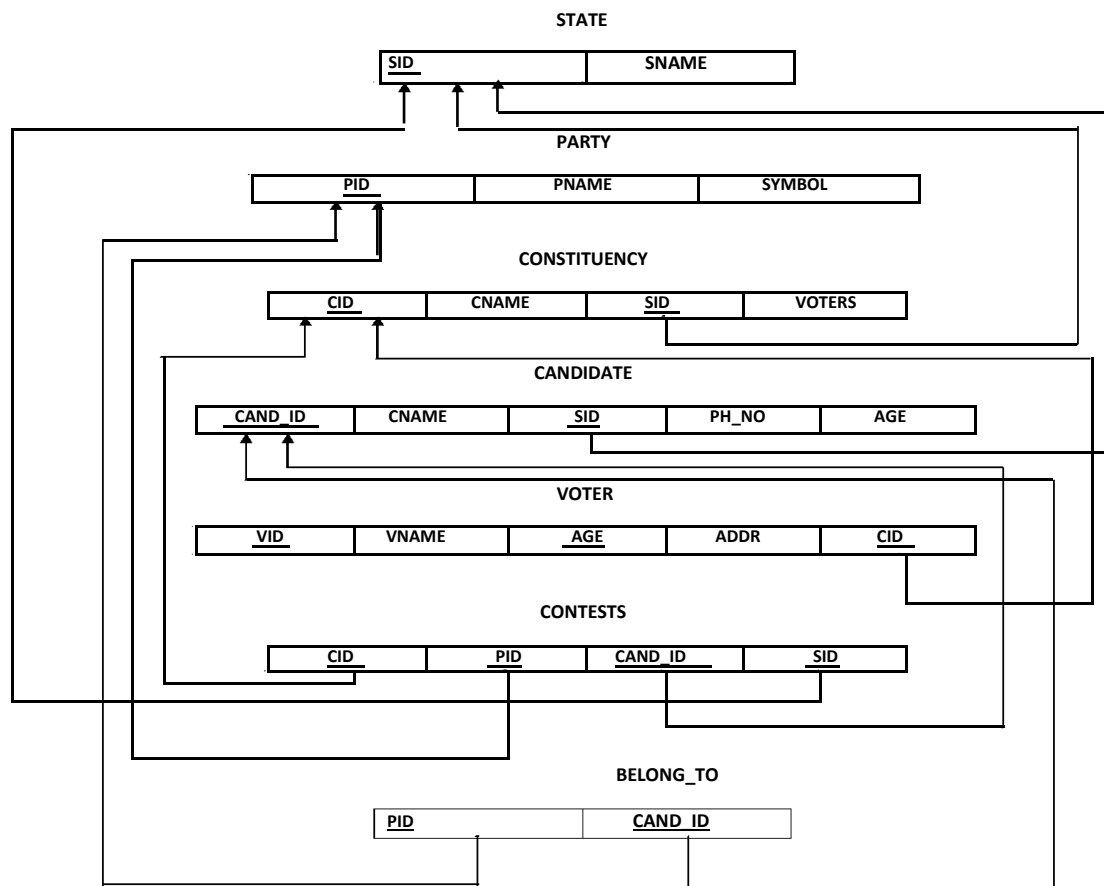


TABLE 1 : STATE_VOTE

```
create table STATE_VOTE(SID varchar(15) primary key, SNAME varchar(25) not
null,UNIQUE(SNAME));
```

TABLE CREATED.

```
insert into STATE_VOTE values('s1','SNAME1');
```

```
select * from STATE_VOTE;
```

SID	SNAME
----	-----
s1	sname1
s2	sname2
s3	sname3
s4	sname5

4 ROWS SELECTED.

.....

TABLE 2 : PARTY

```
create table PARTY(PID varchar(15) primary key,PNAME varchar(25) not null,SYMBOL
varchar(15) not null,unique(SYMBOL));
```

TABLE CREATED.

```
insert into PARTY values('p1','pname1','s1');
```

```
select * from PARTY;
```

PID	PNAME	SYMBOL
----	-----	-----
p1	pname1	s1
p2	pname2	s2
p3	pname3	s3
p4	pname4	s4

4 ROWS SELECTED.

TABLE 3 : CONS

create table CONS(CID varchar(15) primary key, CNAME varchar(15) not null,SID
varchar(15) references STATE_VOTE,VOTERS int not null);

TABLE CREATED.

insert into CONS values('c1','cn1','s1','10');

select * from CONS;

CID	CNAME	SID	VOTERS
----	-----	-----	-----
c1	cn1	s1	10
c2	cn2	s2	20
c3	cn3	s3	25
c4	cn4	s1	15
c5	cn5	s4	12
c6	cn6	s4	18

6 ROWS SELECTED.

.....

TABLE 4 : VOTER

create table VOTER(VID varchar(15) primary key, VNAME varchar(25) not null,AGE int
not null,check(AGE>17), ADDR varchar(30) not null,CID varchar(15) references CONS);

TABLE CREATED.

insert into VOTER values('v1','vn1','19',102,c1,s1,123,'c1');

select * from VOTER;

VID	VNAME	AGE	ADDR	CID
----	-----	-----	-----	-----
v1	vn1	19	102,c1,s1,123	c1
v2	vn2	18	103,c2,s2,124	c2
v3	vn3	20	104,c3,s3,125	c3
v4	vn4	40	105,c4,s4,126	c4
v5	vn5	35	106,c5,s2,127	c5
v6	vn6	38	107,c6,s3,128	c6
v7	vn7	63	108,c5,s2,129	c5

7 ROWS SELECTED.

.....

TABLE 2 : CANDIDATE

```
create table CANDIDATE(CAND_ID varchar(15) primary key ,CNAME varchar(15) not null,PH_NO
int,UNIQUE(PH_NO), AGE int check(AGE>21),SID varchar(15) references STATE_VOTE);
```

TABLE CREATED.

```
insert into CANDIDATE values('cd1','n1','1234','28','s1');
```

```
select * from CANDIDATE;
```

CAND_ID	CNAME	PH_NO	AGE	SID
cd1	n1	1234	28	s1
cd2	n2	3421	25	s2
cd3	n3	5423	38	s3
cd4	n4	7253	68	s4
cd5	n5	1432	45	s1

5 ROW SELECTED.

TABLE 2 : BELONG_TO

```
create table BELONG_TO(CAND_ID varchar(15) references CANDIDATE,  
PID varchar(15) references PARTY);
```

TABLE CREATED.

```
insert into BELONG_TO values('CD1','P1');
```

```
select * from BELONG TO;
```

```
CAND_ID      PID
-----
cd1          p1
cd2          p2
cd3          p3
cd4          p4
cd5          p1
5 ROWS SELECTED.
```

.....

TABLE 7 : CONTESTS

create table CONTESTS(CID varchar(15) references CONS, PID varchar(15) references PARTY,CAND_ID
varchar(15) references CANDIDATE,SID varchar(15) references STATE_VOTE);

TABLE CREATED.

insert into CONTESTS values('c1','p1','cd1','s1');

select * from CONTESTS;

CID	PID	CAND_ID	SID
-----	-----	-----	-----
c1	p1	cd1	s1
c2	p1	cd1	s1
c1	p2	cd2	s2
c2	p3	cd3	s3
c3	p1	cd1	s4
c4	p4	cd4	s3

6 ROWS SELECTED.

QUERIES :

1. List the details of the candidates who are contesting from more than one constituencies which are belongs to different states.

Solution :

SELECT *
FROM CANDIDATE
WHERE CAND_ID IN(SELECT CAND_ID FROM CONTESTS GROUP BY CAND_ID HAVING
COUNT(*)>1);

RESULT :

CAND_ID CNAME PH_NO AGE SID -----
----- cd1 n1 1234 28 s1

2. Display the state name having maximum number of constituencies.

Solution :

```
SELECT *
FROM STATE_VOTE
WHERE SID IN(SELECT SID FROM CONTESTS
              GROUP BY SID HAVING COUNT(*)=(SELECT MAX(COUNT(*))
              FROM CONTESTS GROUP BY SID));
```

RESULT :

SID	SNAME
s1	sname1
s3	sname3

3. Create a stored procedure to insert the tuple into the voter table by checking the voter age.

If voter's age is at least 18 years old, then insert the tuple into the voter else display the "Not an eligible voter msg"

Solution :

STEP 1 : CREATE PROCEDURE:

```
CREATE PROCEDURE INSERT_VOTER(VID VARCHAR) IS BEGIN
    INSERT INTO VOTER VALUES(VID,'VN8',63,'108,C7,S1,129','C6');
END;
/
Procedure Succesfully Completed.
```

STEP 2 : EXECUTE SQL STATEMENT:

```
EXECUTE INSERT_VOTER('V8');
PL/SQL procedure successfully completed.
```

STEP 3 :

```
SELECT * FROM VOTER;
```

VID	VNAME	AGE	ADDR	CID
v1	vn1	19	102,c1,s1,123	c1
v2	vn2	18	103,c2,s2,124	c2
v3	vn3	20	104,c3,s3,125	c3
v4	vn4	40	105,c4,s4,126	c4
v5	vn5	35	106,c5,s2,127	c5
v6	vn6	38	107,c6,s3,128	c6
v7	vn7	63	108,c5,s2,129	c5
v8	vn8	63	108,c7,s1,129	c6

8 ROWS SELECTED.

4. Create a stored procedure to display the number_of_voters in the specified constituency. Where the constituency name is passed as an argument to the stored procedure.

Solution :

STEP 1 : CREATE A RESULTANT TABLE :

```
CREATE TABLE T AS SELECT CNAME,0 AS TOTAL FROM CONS;
```

STEP 2 : CREATE PROCEDURE :

```
CREATE PROCEDURE DISPLAY(CNAME1 VARCHAR) IS
BEGIN
    UPDATE T
        SET TOTAL=(SELECT VOTERS FROM CONS WHERE CNAME=CNAME1)
        WHERE CNAME=CNAME1;
END;
```

STEP 3 : EXECUTE SQL STATEMENT :

```
EXECUTE DISPLAY('CN1');
```

PL/SQL procedure successfully completed.

STEP 4 :

RESULT :

```
SELECT * FROM T;
CNAME      TOTAL
-----
cn1         10
cn2          0
cn3          0
cn4          0
cn5          0
cn6          0
6 rows selected.
```

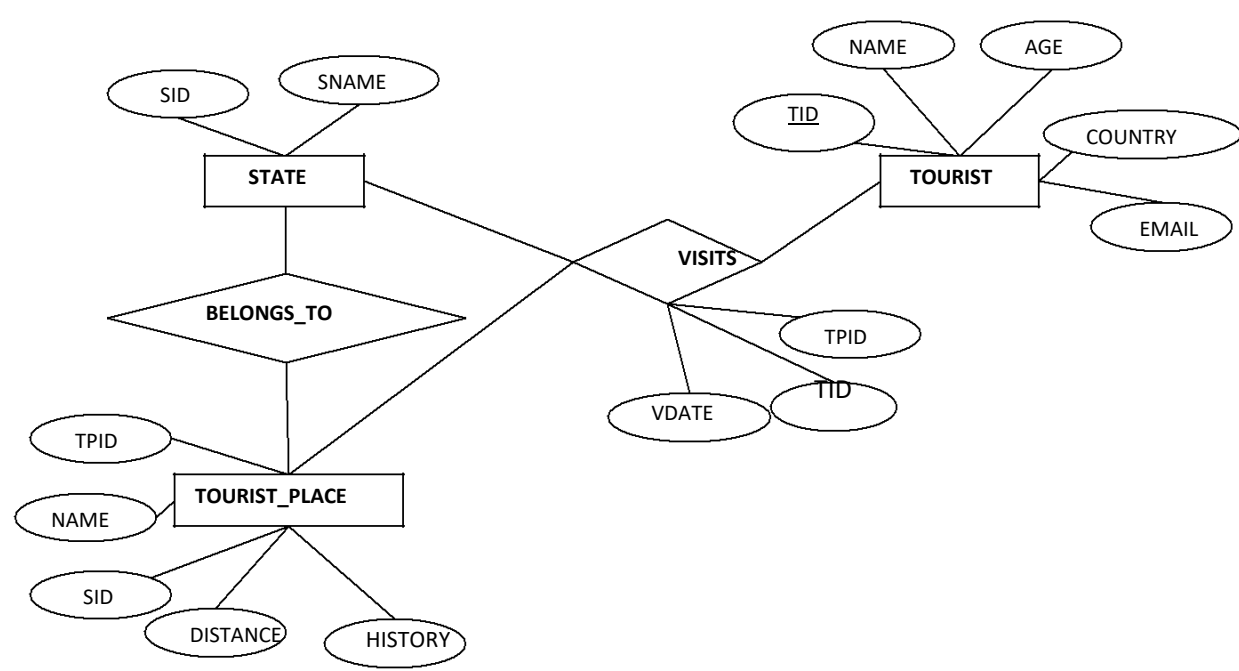
5. Design an ER-diagram for the following scenario, Convert the same into a relational model, normalize Relations into a suitable Normal form and then solve the following queries.

A country can have many Tourist places . Each Tourist place is identified by using tourist_place_id, having a name, belongs to a state, Number of kilometers away from the capital city of that state,history. There are many Tourists visits tourist places every year. Each tourist is identified uniquely by using Tourist_id, having a Name, age, Country and multiple emailids. A tourist visits many Tourist places, it is also required to record the visted_date in the database. A tourist can visit a Tourist place many times at different dates. A Tourist place can be visited by many tourists either in the same date or at different dates.

Queries:

1. List the state name which is having maximum number of tourist places.
2. List details of Tourist place where maximum number of tourists visited.
3. List the details of tourists visited all tourist places of the state “KARNATAKA”.
4. Display the details of the tourists visited at least one tourist place of the state, but visited all states tourist places.
5. Display the details of the tourist place visited by the tourists of all country.

ER DIAGRAM :



SCHEMA DIAGRAM :

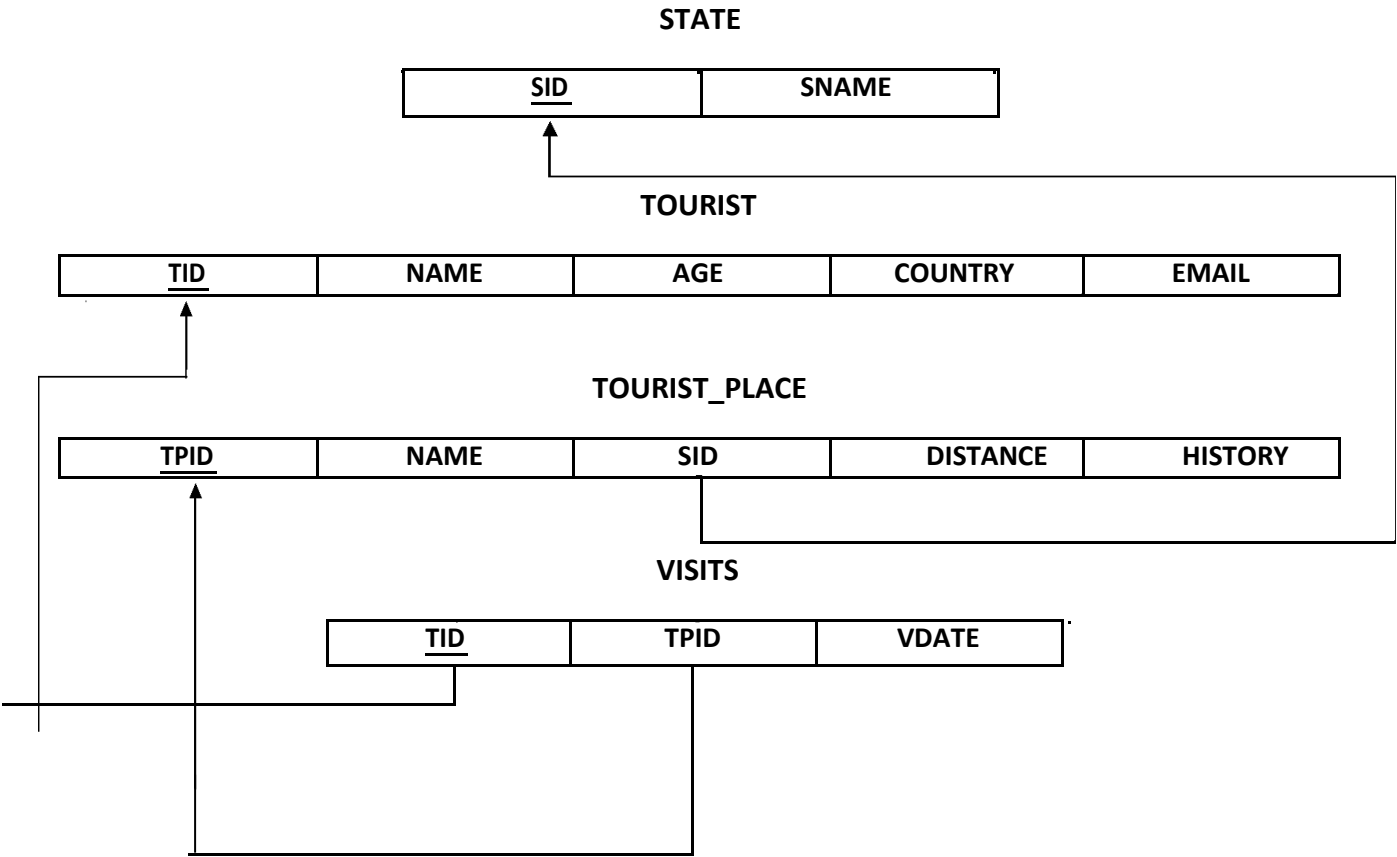


TABLE 1 : STATE

```
create table STATE(SID varchar(15) primary key,SNAME varchar(25) not null);
```

```
TABLE CREATED.
```

```
insert into STATE values('s1','sname1');
```

```
SELECT * FROM STATE;
```

```
SID  SNAME
```

```
----
```

```
s1  sname1
```

```
s2  karnataka
```

```
s3  sname3
```

```
s4  sname4
```

```
s5  sname5
```

```
5 ROWS SELECTED.
```

```
.....
```

TABLE 2 : TOURIST

```
create table TOURIST(TID varchar(15) primary key,NAME varchar(20) not null,AGE int not null,COUNTRY varchar(25) not null,EMAIL varchar(25) not null);
```

```
TABLE CREATED.
```

```
insert into TOURIST values('t1','tname1','20','c1','e1');
```

```
SELECT * FROM TOURIST;
```

```
TID  NAME  AGE  COUNTRY  EMAIL
```

```
-----
```

```
t1    tname1   20    c1    e1@gmail.com
```

```
t2    tname2   18    c2    e2@gmail.com
```

```
t3    tname3   25    c1    e3@gmail.com
```

```
t4    tname4   30    c2    e4@gmail.com
```

```
t5    tname5   24    c3    e5@gmail.com
```

```
t6    tname6   27    c4    e6@gmail.com
```

```
t7    tname7   28    c5    e7@gmail.com
```

```
7 ROWS SELECTED.
```

```
.....
```

TABLE 3 : TOURIST_PLACE

create table TOURIST_PLACE(TPID varchar(25) primary key,NAME varchar(25) not null,SID varchar(15) references STATE,DISTANCE int, check(DISTANCE between 1 and 100),HISTORY varchar(10) not null);

TABLE CREATED.

insert into TOURIST_PLACE values('tp1','karnataka','s1','60','Adilsha');

SELECT * FROM TOURIST_PLACE;

TPID	NAME	SID	DISTANCE	HISTORY
tp1	karnataka	s1	60	Adilsha
tp2	name1	s2	50	Vijayanagar
tp3	name3	s3	55	Mysore
tp4	name4	s4	45	Agra
tp5	name5	s5	35	Kanyakumari
tp6	name6	s5	40	Rameshwar
tp7	name7	s4	56	Ayodhya
tp8	name8	s2	64	Madurai

8 ROWS SELECTED.

TABLE 4 : VISITS

create table VISITS(TPID varchar(25) references TOURIST_PLACE,TID varchar(15) references TOURIST,VDATE DATE not null);

TABLE CREATED.

insert into VISITS values('&TPID','&TID','&VDATE');

SELECT * FROM VISITS;

TPID	TID	VDATE
tp2	t2	21-NOV-21
tp3	t3	22-NOV-21
tp4	t4	23-NOV-21
tp5	t5	24-NOV-21
tp2	t1	25-NOV-21
tp1	t2	26-NOV-21
tp1	t3	26-NOV-21

8 ROWS SELECTED.

.....

QUERIES :

1. List the state name which is having maximum number of tourist places.

Solution :

```
select sname from state where sid in(select sid from tourist_place group by
sid having count(*)>=(select max(count(*))
from tourist_place group by sid));
```

RESULT:

SNAME

sname1

.....

2. List details of Tourist place where maximum number of tourists visited. Solution :

```
select * from tourist_place where tpid in(select tpid from visits
group by tpid
having count(*)>=(select max(count(*))
from visits group by tpid));
```

RESULT :

TPID	NAME	SID	DISTANCE	HISTORY
------	------	-----	----------	---------

tp1

karnataka

s1

60

Adilsha

.....

3. List the details of tourists visited all tourist places of the state "KARNATAKA".

Solution :

```
select * from tourist where tid in(select tid
from visits where tpid in(select tpid
from tourist_place
where sid in(select sid from state
where sname='karnataka'))
order by tid;
```

RESULT :

TID	NAME	AGE	COUNTRY	EMAIL
-----	------	-----	---------	-------

t2

tname2

18

c2

e2@gmail.com

.....

4. Display the details of the tourists visited at least one tourist place of the state, but visited all states tourist places.

Solution :

```
select *  
from tourist  
where tid in(select distinct(tid) from visits  
             where exists(select tid from tourist));
```

TID	NAME	AGE	COUNTRY	EMAIL
-----	-----	-----	-----	-----
t1	tname1	20	c1	e1@gmail.com
t2	tname2	18	c2	e2@gmail.com
t3	tname3	25	c1	e3@gmail.com
t4	tname4	30	c2	e4@gmail.com
t5	tname5	24	c3	e5@gmail.com
t6	tname6	27	c4	e6@gmail.com

.....

5. Display the details of the tourist place visited by the tourists of all country. Solution :

```
select * from tourist_place  
where tpid in(select distinct(tpid)  
from visits where exists(select tpid from tourist_place));
```

TPID	NAME	SID	DISTANCE	HISTORY
-----	-----	-----	-----	-----
tp1	karnataka	s1	60	Adilsha
tp2	name1	s2	50	Vijayanagar
tp3	name3	s3	55	Mysore
tp4	name4	s4	45	Agra
tp5	name5	s5	35	Kanyakumari

9. Consider the following database of student enrollment in courses and books adopted for each course.

STUDENT (regno#: string, name: string, major: string, bdate: date)

COURSE (course#: int, cname: string, dept: String)

TEXT (book ISBN#: int, book_title: string, publisher: string, author: string)

ENROLL (regno#: string, course#: int, sem: int, marks: int)

BOOK_ADOPTION (course#: int, sem: int, book ISBN: int)

Create the above tables by properly specifying the primary keys and the foreign keys

Enter at least 7 to 10 records to each table. Execute SQL queries for the following requirements:

- 1) List out the student details, and their course details. The records should be ordered in a semester wise manner.
- 2) List out the student details under a particular department whose name is ordered in a semester wise
- 3) List out all the book details under a particular course
- 4) Find out the Courses in which number of students studying will be more than 2.
- 5) Find out the Publisher who has published more than 2 books.
- 6) Find out the authors who have written book for I semester, computer science course.
- 7) List out the student details whose total number of months starting from their date of birth is more than 225
- 8) Find out the course name to which maximum number of students have joined

Table: Student

Create table student (
regno varchar2(5) primary key,
name varchar2(10),
major varchar2(10),
bdate date);

insert into student values('®no','&name','&major',&bdate);

Select * from student;

regno	name	major	Bdate
mca01	sunil	a	1-june-2020
mca02	anil	b	1-july-2020
mca03	ajay	c	5-mar-2020
mca04	abhi	d	10-june-2020
mca06	sushma	e	10-may-2020
mca05	aishwarya	f	4-june-2020
mca07	shreya	g	10-11-2020

Table: Course

```
Create table course(  
courseid number(4) primary key,  
cname varchar2(10),  
dept varchar2(10));
```

```
Insert into course values(&courseid, '&cname', '&dept');
```

```
Select * from course;
```

courseid	cname	dept
1	dbms	mca
2	os	mca
3	cn	mca
4	unix	cs
5	se	cs
6	ai	aiml
7	ml	aiml

Table: Text

```
create table text(book_ISBN number(4) primary key,  
book_title varchar2(10),  
publisher varchar2(10),  
author varchar2(10));
```

```
insert into text values(&book_ISBN, '&book_title', '&publisher', '&author');
```

```
Select * from text;
```

<u>book_ISBN</u>	book_title	publisher	author
123	c	tata	a1
456	c++	willey	a2
111	dbms	tata	a3
112	unix	tata	a4
189	os	willey	a5
198	IoT	newage	a6
899	java	cengage	a7

Table:enroll

```
create table enroll( regno varchar2(5) references student(regno),  
courseid number(4) references course(courseid),  
sem number(4),  
marks number(4));
```

```
insert into enroll values('&regno','&courseid',&sem,&marks);
```

```
select * from enroll
```

regno	courseid	sem	marks
mca01	1	2	50
mca02	2	1	60
mca03	3	1	86
mca04	4	2	85
mca05	5	2	99
mca06	6	2	88
mca07	6	2	83

Table: Book_Adoption

```
create table book_adoption( courseid number(4) references course(courseid),
sem number(4),
book_ISBN number(4) references text(book_ISBN));
```

```
insert into book_adoption values(&courseid,&sem,&book_ISBN);
```

```
select * from book_adoption;
```

courseid	sem	book_ISBN
1	2	123
2	1	456
4	2	111
3	1	112
5	2	189
6	2	198
7	2	899

1) List out the student details, and their course details. The records should be ordered in a semester wise manner.

```
select s.regno,s.name,s.bdate,e.courseid,e.sem,e.marks
from student s,enroll e
where s.regno=e.regno order by sem
```

```
SQL> /
```

REGNO	NAME	BDATE	COURSEID	SEM	MARKS
-----	-----	-----	-----	-----	-----
mca02	anil	01-JUL-20	2	1	60
mca03	ajay	05-MAR-20	3	1	86
mca05	aishwarya	04-JUN-20	5	2	99
mca06	sushma	10-MAY-20	6	2	88
mca07	shreya	10-NOV-20	6	2	83
mcao1	sunil	01-JUN-20	1	2	50
mca04	abhi	10-JUN-20	4	2	85

7 rows selected.

2) List out the student details under a particular department whose name is ordered in a semester wise

```
select s.regno,s.name,s.bdate,e.courseid,e.sem,e.marks,c.dept
from student s,enroll e,course c
where s.regno=e.regno and c.courseid=e.courseid and c.dept='mca'
order by sem
```

SQL> /

REGNO	NAME	BDATE	COURSEID	SEM	MARKS	DEPT
mca02	anil	01-JUL-20	2	1	60	mca
mca03	ajay	05-MAR-20	3	1	86	mca
mca01	sunil	01-JUN-20	1	2	50	mca

3) List out all the book details under a particular course

```
select t.book_ISBN,t.book_title,t.publisher,t.author,b.courseid
from text t,book_adoption b
where t.book_ISBN=b.book_ISBN and b.courseid=1
```

SQL> /

BOOK_ISBN	BOOK_TITLE	PUBLISHER	AUTHOR	COURSEID
123	c	tata	a1	1

```
select t.book_ISBN,t.book_title,t.publisher,t.author,b.courseid
from text t,book_adoption b
where t.book_ISBN=b.book_ISBN and courseid in
(select courseid from course where cname='dbms')
```

SQL> /

BOOK_ISBN	BOOK_TITLE	PUBLISHER	AUTHOR	COURSEID
123	c	tata	a1	1

4) Find out the Courses in which number of students studying will be more than 2.

```
select cname,courseid from course where courseid in
(select courseid from enroll group by courseid having count(*) >=2)
```

SQL> /

CNAME	COURSEID
ai	6

5) Find out the Publisher who has published more than 2 books.

select publisher from text group by publisher having count(*) >=2

SQL> /

PUBLISHER

willey

tata

6) Find out the authors who have written book for I semester, computer science course.

select * from text where book_ISBN in
 (select book_ISBN from book_adoption
 where sem=1 and courseid in
 (select courseid from course where cname='dbms'))

SQL> /

BOOK_ISBN BOOK_TITLE PUBLISHER AUTHOR

123 c tata a1

select * from text where book_ISBN in
 (select book_ISBN from book_adoption
 where sem=1 and courseid in
 (select courseid from course where dept='mca'))

SQL> /

BOOK_ISBN BOOK_TITLE PUBLISHER AUTHOR

456 c++ willey a2

112 unix tata a4

7) List out the student details whose total number of months starting from their date of birth is more than 225

select months_between(current_date,bdate) as No_months from student;

SQL> /

NO_MONTHS

36.2777494

35.2777494

39.1487171

35.9874268

36.1809752

36.9874268

30.9874268

```
select * from student where months_between(current_date,bdate) > 35;
```

SQL> /

REGNO	NAME	MAJOR	BDATE
mcao1	sunil	a	01-JUN-20
mca02	anil	b	01-JUL-20
mca03	ajay	c	05-MAR-20
mca04	abhi	d	10-JUN-20
mca05	aishwarya	f	04-JUN-20
mca06	sushma	e	10-MAY-20

6 rows selected.

8) Find out the course name to which maximum numbers of students have joined

```
select * from course where courseid in
    (select courseid from enroll group by courseid
     having count(*)=(select max(count(*))
                      from enroll group by courseid))
```

SQL> /

COURSEID	CNAME	DEPT
6	ai	aiml

6.Create an XHTML page that provides information about your department. Your XHTML page must use the following tags: a) Text Formatting tags b) Horizontal rule c) Meta element d) Links e) Images f) Tables (Use of additional tags encouraged).

College.html

```
<?xml version="1.0" encoding="utf-8"?>
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.1//EN"
"http://www.w3.org/TR/xhtml11/DTD/xhtml11.dtd">
<html xmlns="http://www.w3.org/1999/xhtml">
<head>
<title>Home Page</title>
<Meta name="Author" content=" S S Adhyapak"/>
</head>
<body>
<h1><center><b> BLDEA's DR PG HALAKATTI COLLEGE OF ENGINEERING & TECHNOLOGY,
VIJAYPUR</b></center></h1>
<h2><p><center>WELCOME to Post Graduate Section</section></center></h2> <center></center>
<a href="mca.html">MCA</a>
<br/>
<a href="course.html">Course Details</a>
<p>
</body>
</html>
```

Course.html

```
<?xml version="1.0" encoding="utf-8"?>
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.1//EN"
"http://www.w3.org/TR/xhtml11/DTD/xhtml11.dtd">
<html xmlns="http://www.w3.org/1999/xhtml">
<head>
<title>Home page</title>
</head>
<body>
<h1><b>Total no. of students in MCA</b></h1>
<br/>
<hr/>
<ol>
<li><a href="firstyear.html">First year(56)</a></li>
<li>Second Year(46)</li>
<li>Second Year(46)</li>
</ol>
<br/>
<hr/>
<p> PLease contact for MCA department for more details</p> <br/>
<a href="college.html">Home</a>
</body>
</html>
```

Firstyear.html

```
<?xml version ="1.0" encoding="utf-8"?>
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.1//EN"
"http://www.w3.org/TR/xhtml11/DTD/xhtml11.dtd"> <html xmlns="http://www.w3.org/1999/xhtml">
<head>
<title>Home page</title>
</head>
<body>
<h1><b>Total no. of students in MCA First Year</b><hr/> <br/>
<ul>
<li>Girls 30</li>
<li>Boys 26</li>
</ul>
<br/>
<hr/>
<p> Please contact for MCA department for more details</p> <br/>
<a href="college.html">Home</a>
</body>
</html>
```

Mca.html

```
<?xml version ="1.0" encoding="utf-8"?>
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.1//EN"
"http://www.w3.org/TR/xhtml11/DTD/xhtml11.dtd"> <html xmlns="http://www.w3.org/1999/xhtml">
<head>
<title>Lab program 1</title>
</head>
<body>
<h1><center> Master Of Computer Applications</h1></center> <center><p> Master Of Computer
Application</center></p>
<center><table border="5" width="20%"> <!--table with name & email--> <tr>
<th width="240">Duration</th>
<th>INTAKE</th>
</tr>
<tr>
<td width="240">2 Years</td>
<td>12</td>
</tr>
<tr>
<td width="240">3 Years</td>
<td>60</td>
</tr>
</table>
</center>
<hr>
<p>
<em><b>Please contact MCA department for More details</b></em> </p>
<a href="college.html">Home</a>
</body>
</html>
```

Output

**BLDEA's DR PG HALAKATTI
COLLEGE OF ENGINEERING & TECHNOLOGY,VIJAYPUR
WELCOME to Post Graduate Section**



Course.html

Total no. of students in MCA

1. First year(56)
2. Second Year(46)
3. Second Year(46)

PLease contact for MCA department for more details

Home

Firstyear.html

Total no. of students in MCA First Year

- **Girls 30**
- **Boys 26**

Please contact for MCA department for more details

Home

7. Develop and demonstrate a XHTML file that includes JavaScript script for the following problems:

a) Input: A number n obtained using prompt Output: The first n Fibonacci numbers

b) Input: A number n obtained using prompt Output: A table of numbers from 1 to n and their squares using alert

Prog7a.html

```
<? xml version="1.0" encoding="utf-8"?>
<! DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.1 //EN"
http://www.w3.org/TR/Xhtml11/DTD/xhtml11.dtd">
<html xmlns="http://www.w3.org/1999/xhtml">
<head>
<title>Fibonacci numbers</title>
</head>
<body>
<script type="text/javascript">
var fib1=0,fib2=1,fib=0;
var num=prompt("enter a number:\n","");
if (num!=null&&num>0)
{
document.write("<h1>"+num+"Fibonacci are<br></h1>");
if(num==1)
document.write("<h1>"+fib1+"</h1>");
else
document.write("<h1>"+fib1+" "+fib2+"</h1>");
for(i=3;i<=num;i++)
{
fib=fib1+fib2;
document.write("<h1>"+fib+"</h1>");
fib1=fib2;
fib2=fib;
}
}
else
alert("No proper input");
</script>
</body>
</html>
```

Prog7 output

5
Fibonacci are
0
1
1
2
3

Prog7b.html

```
<? xml version="1.0" encoding="utf-8"?>

<! DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.1 //EN""
http://www.w3.org/TR/Xhtml11/DTD/xhtml11.dtd"> <html xmlns="http://www.w3.org/1999/xhtml">
<body>

<script type="text/javascript">

var num=prompt("enter a number:\n","");

if(num>0&&num!=null){

msgstr="Number and its squares are\n";

for(i=1;i<=num;i++)

{

msgstr=msgstr+i+"-"+i*i+"\n";

}

alert(msgstr)

}

else

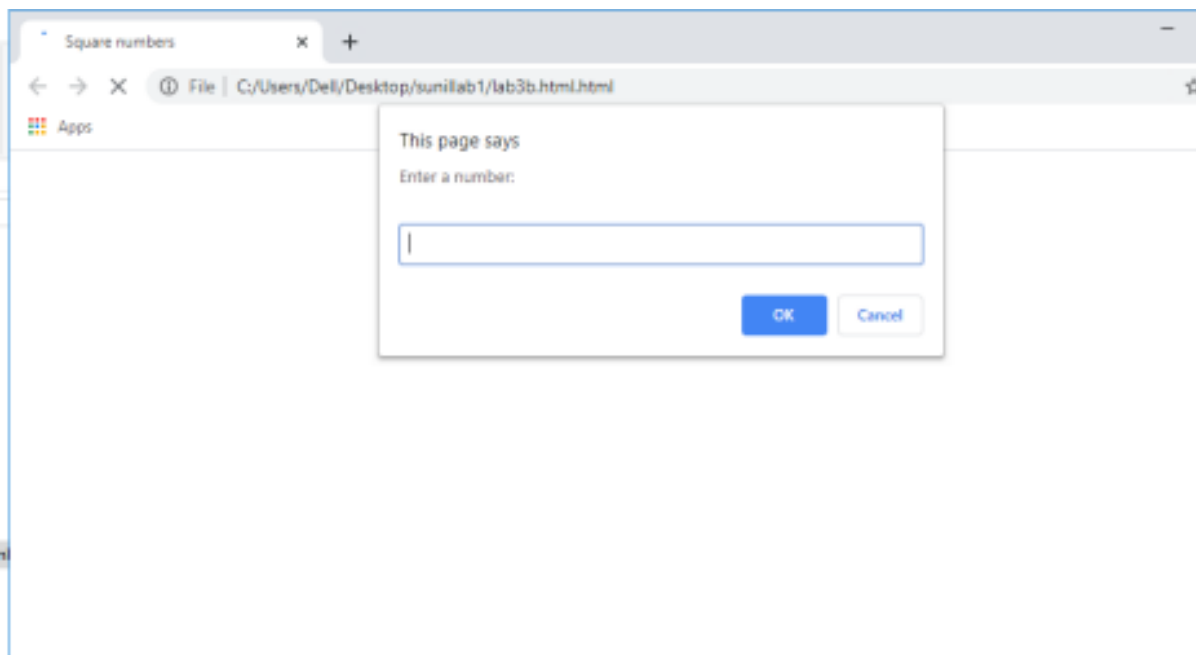
alert("no input supplied");

</script>

</body>

</html>
```

Prog7b output





8. Develop and demonstrate, using JavaScript script, a XHTML document that contains three short paragraphs of text, stacked on top of each other, with only enough of each showing so that the mouse cursor can be placed over some part of them. When the cursor is placed over the exposed part of any paragraph, it should rise to the top to become completely visible. Modify the above document so that when a text is moved from the top stacking position, it returns to its original position rather than to the bottom.

Prog 8a.html

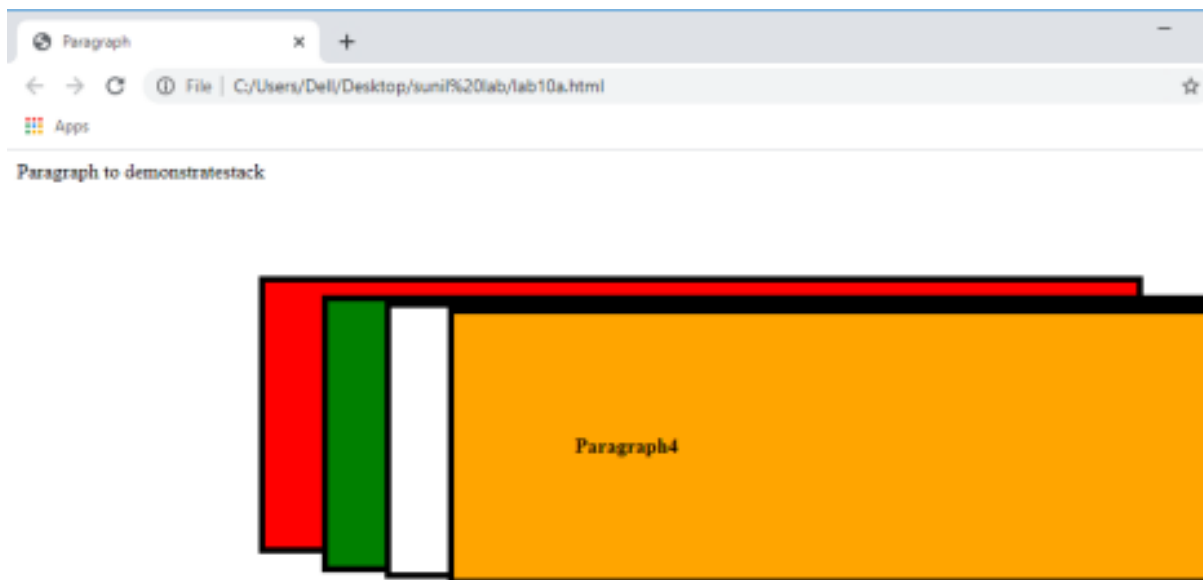
```
<?xml version="1.0" encoding="utf-8"?>
<!DOCTYPE html PUBLIC "-//w3c//DTD XHTML 1.1//EN"
"http://www.w3.org/TR/xhtml11.dtd">
<html xmlns="http://www.w3.org/1999/xhtml">
<head>
<title>Paragraph</title>
<link rel="stylesheet" type="text/css" href="prog10a.css"/>
<script type="text/javascript">
var toplayer="layer4";
function stack(toTop)
{
var oldtop=document.getElementById(toplayer).style;
var newtop=document.getElementById(toTop).style;
oldtop.zIndex="0";
newtop.zIndex="10";
toplayer=toTop;
}
</script>
</head>
<body>
<div class="ls1" id="layer1" onmouseover="stack('layer1')">
<b>Paragraph1</b></div>
<div class="ls2" id="layer2" onmouseover="stack('layer2')">
<b>Paragraph2</b></div>
<div class="ls3" id="layer3" onmouseover="stack('layer3')">
<b>Paragraph3</b></div>
<div class="ls4" id="layer4" onmouseover="stack('layer4')">
<b>Paragraph4</b></div>
</body>
</html>
```

prog10a.css

```
.ls1 {border:solid thick black; padding:1in;
width:500px;
background-color:red; position:absolute;
top:100px;
left:200px;
}
.ls2 {border:solid thick black; padding:1in;
width:500px;
background-color:green; position:absolute;
top:115px;
left:250px;
}
```

```
.ls3 {border:solid thick black; padding:1in;
width:500px;
background-color:white; position:absolute;
top:120px;
left:300px;
}
.ls4 {border:solid thick black; padding:1in;
width:500px;
background-color:orange; position:absolute;
top:125px;
left:350px;
}
```

Output



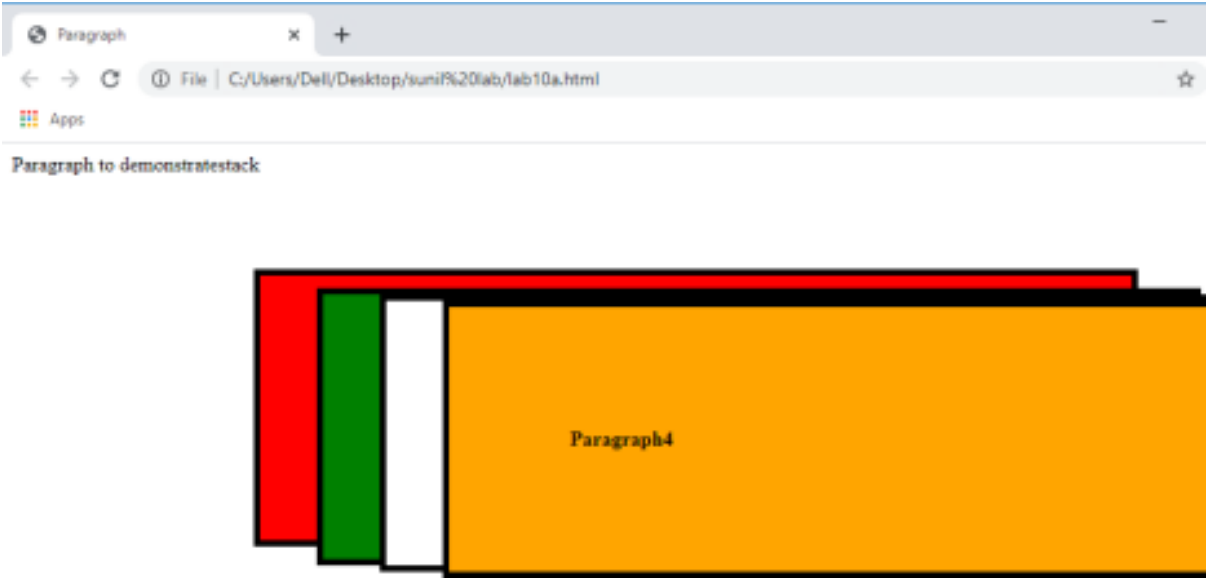
Prog 8b.html

```
<?xml version="1.0" encoding="utf-8"?>
<!DOCTYPE html PUBLIC "-//w3c//DTD XHTML 1.1//EN" "http://www.w3.org/TR/xhtml11.dtd">
<html xmlns="http://www.w3.org/1999/xhtml"> <head>
<title>Paragraph</title>
<link rel="stylesheet" type="text/css" href="prog10b.css"/> <script type="text/javascript">
var toplayer="layer4";
var origpos;
function stack(toTop)
{
var newtop=document.getElementById(toTop).style; newtop.zIndex="10";
toplayer=toTop;
}
</script>
</head>
<body>
Paragraph to demonstratestack
<div class="ls1" id="layer1" onmouseover="stack('layer1')"> <b>Paragraph1</b></div>
<div class="ls2" id="layer2" onmouseover="stack('layer2')"> <b>Paragraph2</b></div>
<div class="ls3" id="layer3" onmouseover="stack('layer3')"> <b>Paragraph3</b></div>
<div class="ls4" id="layer4" onmouseover="stack('layer4')"> <b>Paragraph4</b></div>
</body>
</html>
```

Prog10b.css

```
.ls1 {border:solid thick black; padding:1in;
width:500px;
background-color:red; position:absolute;
top:100px;
left:200px;
}
.ls2 {border:solid thick black; padding:1in;
width:500px;
background-color:green; position:absolute;
top:115px;
left:250px;
}
.ls3 {border:solid thick black; padding:1in;
width:500px;
background-color:white; position:absolute;
top:120px;
left:300px;
}
.ls4 {border:solid thick black; padding:1in;
width:500px;
background-color:orange; position:absolute;
top:125px;
left:350px;
}
```

Output



10. Develop, test and validate an XHTML document that has checkboxes for apple (59 cents each), orange (49 cents each), and banana (39 cents each) along with submit button. Each check boxes should have its own onclick event handler. These handlers must add the cost of their fruit to a total cost. An event handler for the submit button must produce an alert window with the message, your total cost is \$xxx, where xxx is the total cost of the chose fruit including 5 percent of sales tax. Modify the document to accept quantity for each item using textboxes.

```
<? xml version="1.0" encoding="utf-8"?>
<!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.1 //EN""
http://www.w3.org/TR/Xhtml11/DTD/xhtml11.dtd">
<html xmlns="http://www.w3.org/1999/xhtml">
<head>
<script type="text/javascript" src="check.js">
</script>
</head>
<body>
<h1>Select the following fruits</h1>
<form id="myform" action=" " onsubmit="displayresult()">
Quantities<input type="text" id="tbap"><input type="checkbox" class="chk" id="one"
onclick="updatebox()"/>Apple<br><br>
Quantities<input type="text" id="tbor"><input type="checkbox" class="chk" id="two"
onclick="updatebox()"/>Orange<br><br>
Quantities<input type="text" id="tbbn"><input type="checkbox" class="chk" id="three"
onclick="updatebox()"/>Banana<br><br>
<input type="text" id="tbtb" style="display:none"/>
<p>
<input type="submit" value="submit order"/>
<input type="reset" value="Clear Order Form"/>
</form>
</p>
</body>
</html>
```

Output

Select the folloeing fruits

Quantities1 ☒ Apple

Quantities2 ☒ Orange

Quantities3 ☒ Banana

287.7

submit order Clear Order Form